Natural Language Processing

Coursework 1

A Statistical POS Tagger and PCFG Parser Trained on the Penn Treebank, Tested on Sri Lankan News Text

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1. Introduction

A statistical parser does not decide the grammatical category of a word or the structure of a sentence with hand-written rules. Instead it estimates, from a manually annotated corpus, how probable each category and structure is in context, and it chooses the most probable analysis. The Penn Treebank (PTB) tag set supplies the standard category labels that make these probabilities comparable across words and sentences.

This coursework has three parts:

1. **Task 1 (40%):** build a statistical part-of-speech (POS) tagger and a probabilistic context-free grammar (PCFG) parser from the labelled PTB training data, and present the model statistics (tag-transition, emission and grammar-rule probabilities).
2. **Task 2 (10%):** select 150 sentences from the Daily Mirror (Sri Lanka) online newspaper and tag them manually with the PTB tag set.
3. **Task 3 (50%):** test each of the 150 sentences against the Task 1 model with standard metrics, measure how likely each sentence is under the PTB model, and group the sentences into categories of alignment.

All work is in three Jupyter notebooks, which can be re-run from start to finish: Task1_POS_Tagger_and_PCFG_Model.ipynb, Task2_Tagged_Sentences.ipynb and Task3_Test_Results.ipynb. The PCFG parser data is in the model/ folder and the test results are in the results/ folder.

2. Training Data

2.1 Choice of corpus

The brief points to the Kaggle *Penn Treebank Dataset* (ptb.train.txt, ptb.valid.txt, ptb.test.txt). On inspection, these files are the Mikolov language-modelling version of the PTB: the text is lower-cased, numbers are replaced by N, rare words by <unk>, and punctuation is removed. They contain **no POS tags and no parse trees**, so tag-transition, emission and grammar-rule probabilities cannot be estimated from them.

As the brief instructs ("Use NLTK Penn TreeBank Dataset for downloading – Labeled"), the labelled training data was taken from NLTK's treebank corpus: the Wall Street Journal (WSJ) section of the PTB, files wsj_0001–wsj_0199. It contains both POS-tagged sentences and full parse trees. Following the lecturer's instruction, only this training material was used, and no validation or test data was used to build the model. The notebook can optionally load ptb.train.txt to show that it is untagged and to report vocabulary overlap.

2.2 Corpus statistics

Property	Value
Source files	199 (wsj_0001 – wsj_0199)
Sentences (parse trees)	3,914
Word tokens (excluding -NONE- empty elements)	94,084
Distinct word types	11,968
POS tags observed	45 (the full PTB tag set, including punctuation tags)

Property	Value
Most frequent tags	NN 14.0%, IN 10.5%, NNP 10.0%, DT 8.7%, NNS 6.4%, JJ 6.2%

Table 1. The labelled PTB training data.

Empty elements (traces and null complementisers, tagged -NONE-) were removed before training because they are not pronounced words. Function tags and co-indices on phrase labels were removed for the grammar, so that for example NP-SBJ-1 became NP and PP-LOC became PP.

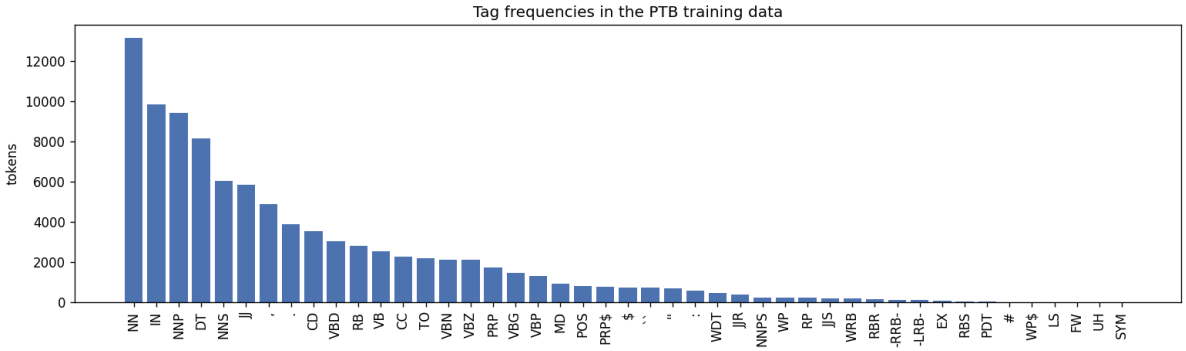


Figure 1. Frequency of each POS tag in the training data.

3. Task 1 – The Statistical POS Tagger and PCFG Parser

The model has two parts that work together. A bigram Hidden Markov Model (HMM) captures word categories in context: which tag follows which, and which words each tag produces. A PCFG induced from the treebank trees captures phrase structure: how tags combine into phrases, and phrases into sentences.

3.1 Bigram HMM tagger

For a sentence $w_1 \ldots w_n$ with tags $t_1 \ldots t_n$, the HMM gives the joint probability

$$P(w_1 \ldots w_n , t_1 \ldots t_n) = \prod_i P(t_i \mid t_{i-1}) \cdot \prod_i P(w_i \mid t_i), \quad \text{with } t_0 = \langle s \rangle \text{ and } t_{n+1} = \langle /s \rangle$$

Transition probabilities were estimated from tag-bigram counts with add-k (Lidstone) smoothing, k = 0.1, so that tag sequences unseen in training still get a small non-zero probability:

$$P(t_i \mid t_{i-1}) = (C(t_{i-1}, t_i) + k) / (C(t_{i-1}) + k \cdot |T|)$$

Emission probabilities were estimated from word–tag counts. Unseen words matter a great deal for Sri Lankan news, which is full of local names. They are handled with *word signatures*: a word's shape (capitalised, contains digits, contains a hyphen) plus its suffix (-ing, -ed, -tion, -ly, -s, …), for example UNK-CAP or UNK-ing. Signature probabilities are estimated from words seen only once in training (hapax legomena), because these behave most like new words. Known words use add-ε smoothing, so a tag never seen with a word is possible but heavily penalised:

$$P(w \mid t) = (C(w, t) + \varepsilon) / (C(t) + \varepsilon \cdot |V|), \quad \varepsilon = 0.001 \quad \text{(known words)}$$

For unseen words the probability is split into how open the tag is to new words and how well the signature fits the tag. H(t) is the number of hapax tokens tagged t:

$$P(w_unk \mid t) = [(H(t) + 0.01) / (C(t) + 1)] \cdot [(H(sig(w), t) + P(sig(w))) / (H(t) + 1)]$$

This way open classes (NN, NNP, JJ, VBN …) receive unseen words, while closed classes (DT, IN, CC, UH, FW …) almost never do. An earlier version, which smoothed each tag's signature distribution independently, let rare tags such as FW and UH attract whole runs of words in the Daily Mirror sentences. Correcting it raised held-out accuracy from 93.1% to 94.7%.

Word lookup tries the exact form first and then the lower-cased form, so a capitalised sentence-initial word is not treated as unknown. Tagging uses the Viterbi algorithm to find the most probable tag sequence.

3.2 The tag-transition matrix

The full 46 × 46 transition matrix (the 45 tags plus the start state as rows, the 45 tags plus the end state as columns) is saved as model/transition_matrix.csv, and the raw counts as model/transition_counts.csv. Figure 2 shows the matrix for the 22 most frequent tags, and Table 2 lists the most probable next tags after some important categories.

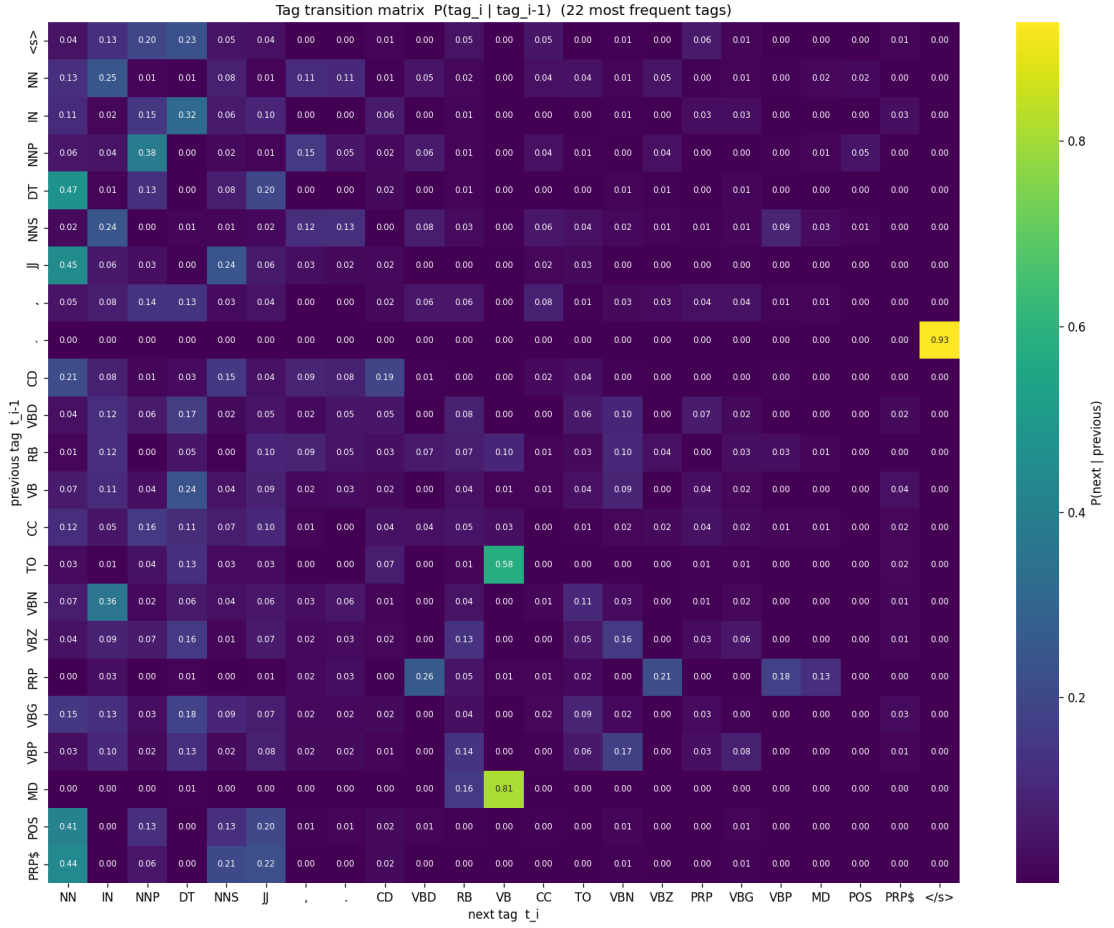


Figure 2. Tag-transition matrix $P(t_i \mid t_{i-1})$ for the 22 most frequent tags (rows: previous tag; columns: next tag).

Previous tag	1st most likely	2nd	3rd
<s>	DT (0.23)	NNP (0.20)	IN (0.13)
DT	NN (0.47)	JJ (0.20)	NNP (0.13)
JJ	NN (0.45)	NNS (0.24)	JJ (0.06)
NN	IN (0.25)	NN (0.13)	, (0.11)
NNP	NNP (0.38)	, (0.15)	VBD (0.06)

Previous tag	1st most likely	2nd	3rd
IN	DT (0.32)	NNP (0.15)	NN (0.11)
MD	VB (0.81)	RB (0.16)	DT (0.01)
TO	VB (0.58)	DT (0.13)	CD (0.07)

Table 2. Most probable next tags. The matrix encodes English word order: modals are followed by base-form verbs (MD → VB 0.81), determiners by nouns or adjectives, and names by names (NNP → NNP 0.38, e.g. multi-word names).

3.3 Emission probabilities

The emission table (model/emission_probs.csv) holds P(word | tag) for all 13,341 word–tag pairs seen in training. For example, P(the | DT) = 0.494 and P(of | IN) = 0.235. Ambiguous words spread their probability over several tags (Figure 3). This ambiguity is what the transition probabilities resolve. The unknown-word model (model/unknown_word_model.csv) assigns, for example, NNP to an unseen capitalised word, NN to an unseen lower-case word, and JJ to an unseen hyphenated -ed word.

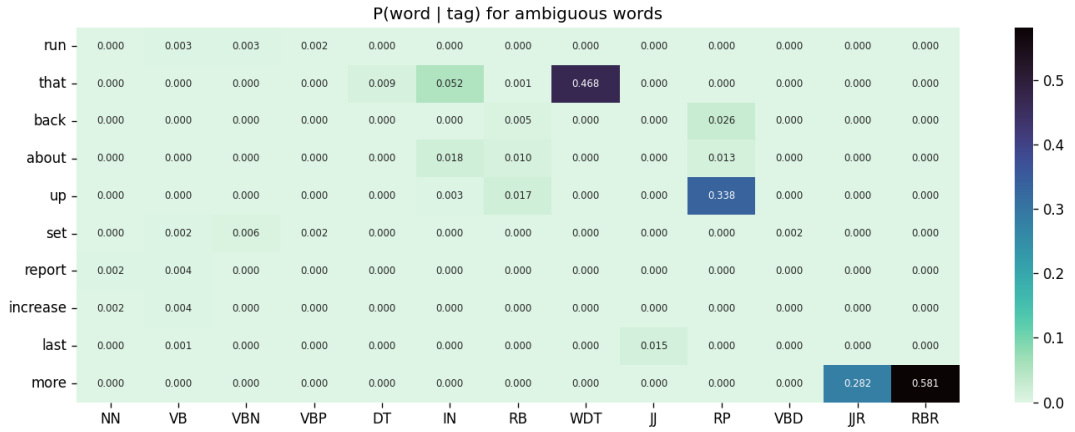


Figure 3. Emission probabilities P(word | tag) for ambiguous words.

3.4 PCFG induction

Two grammars were induced with maximum-likelihood estimation, P(A → β) = C(A → β) / C(A):

- **Lexicalised PCFG** (model/pcfg_grammar_lexicalised.txt, model/pcfg_rules.csv): induced with nltk.induce_pcfg from the cleaned n-ary trees. It has 17,105 rules: 3,764 phrasal rules and 13,341 lexical rules such as NN → 'market'. It uses 73 left-hand-side symbols, and it is the full PCFG parser data.
- **Parser grammar in Chomsky Normal Form** (model/pcfg_parser_grammar_cnf.txt): the terminals are the POS tags. Unary chains are collapsed (e.g. S+VP) and rules are binarised with horizontal Markovisation of order 1 (Klein & Manning, 2003). The result has 4,783 rules and 761 non-terminals. Parsing tag sequences, not words, lets the grammar score the structure of any sentence, even one whose words (e.g. Kurunegala) never occur in the WSJ. Evidence about the words themselves is already covered by the HMM.

LHS	RHS	Count	P(rule)
NP	NP PP	3,507	0.112
NP	DT NN	2,877	0.092
NP	PRP	1,705	0.055
NP	NNS	1,324	0.042
PP	IN NP	7,596	0.815

LHS	RHS	Count	P(rule)
PP	TO NP	826	0.089
PP	IN S	279	0.030
PP	ADVP IN NP	67	0.007
S	NP VP	2,862	0.302
S	VP	2,335	0.247
S	NP VP .	1,761	0.186
S	PP , NP VP .	270	0.029
SBAR	IN S	1,005	0.349
SBAR	S	855	0.297
SBAR	WHNP S	685	0.238
SBAR	WHADVP S	155	0.054
VP	TO VP	1,256	0.087
VP	VB NP	871	0.060
VP	MD VP	756	0.052
VP	VBN PP	547	0.038

Table 3. Most probable expansions of the main phrase categories in the lexicalised PCFG (e.g. PP → IN NP accounts for 81% of all prepositional phrases).

3.5 Probabilistic CKY parsing

A Viterbi CKY parser finds the most probable tree T* over a tag sequence and its log-probability log P(T*, $t_1 \ldots t_n$). To keep long news sentences fast, each chart cell keeps its 40 best constituents (beam pruning). If the grammar cannot build a single tree over the whole sentence, the parser returns a fragment parse: the fewest, most probable constituents that together cover the sentence. Each join the grammar could not make costs the log-probability of the grammar's rarest binary rule. This keeps fragment parses comparable with full parses, and makes the number of fragments a direct measure of structural mismatch.

3.6 Validation of the model

Before the model was applied to new text, its quality was measured on a held-out 10% of the training files (19 files, 320 sentences). The model used for this check was trained on the other 90%. The final model was then retrained on all the training data.

Component	Metric	Result
POS tagger (HMM + Viterbi)	Token accuracy	94.73%
	Accuracy on known words	96.07%
	Accuracy on unknown words (10.0% of tokens)	82.77%
PCFG parser (CKY, gold tags, ≤ 40 tokens)	Coverage (full parse found)	98.36%
	Labelled bracket precision	77.12%
	Labelled bracket recall	73.40%
	Labelled bracket F1 (PARSEVAL)	75.22%

Table 4. Held-out validation of the Task 1 model.

These figures are typical of a bigram HMM and an unlexicalised treebank PCFG trained on about 100,000 tokens. For comparison, trigram taggers trained on the full 1-million-word WSJ reach about 96–97%. They show that the model is a sound reference for judging the Daily Mirror sentences, and they set the baseline against which the Task 3 results are compared.

4. Task 2 – The Daily Mirror Test Sentences

4.1 Selection

150 sentences were taken from 11 recent articles on the Daily Mirror website (www.dailymirror.lk, including its education section). The articles were published around 25–26 September 2026 and cover health, food prices, energy policy, crime (narcotics), a parliamentary debate, examination results, a sponsored education feature, international news (a BBC report republished by the Daily Mirror), a landslide warning, sport, and culture. Their titles, URLs and sentence ranges are listed in task2_sources.txt. Following the lecturer's guidance, very short sentences were avoided: after PTB tokenisation the sentences contain 4,237 tokens, with a mean of 28.2 tokens per sentence (minimum 9, maximum 63).

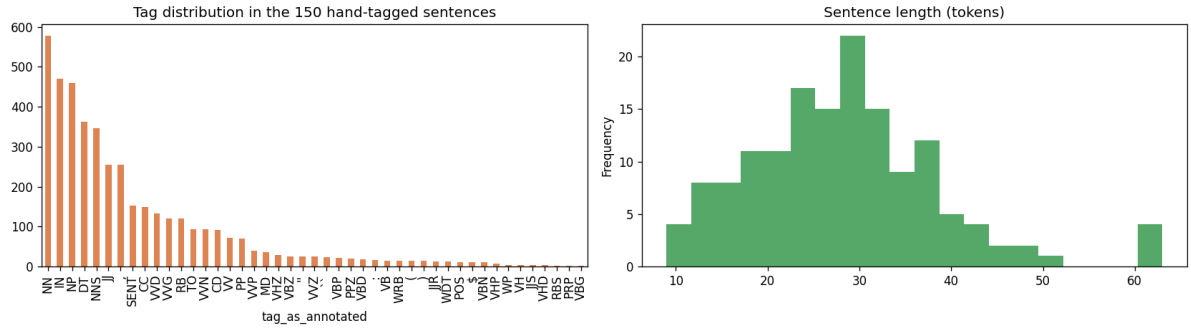


Figure 4. Tag distribution in the 150 hand-tagged sentences (course tag spelling) and sentence lengths in tokens.

4.2 Tokenisation and manual tagging

Each sentence was tokenised with the PTB conventions (NLTK TreebankWordTokenizer) so that its tokens match those of the training data. Clitics are split (won't → wo n't, Lanka's → Lanka 's), punctuation is separated, straight double quotes become `` and '', and brackets become -LRB- and -RRB-. Curly quotes and dashes from the web page were normalised first.

The course tag sheet provided by the lecturer ("Penn-TreeBank" tab) uses the TreeTagger spelling of the Penn Treebank tags: NP and NPS for proper nouns, PP and PPZ for pronouns, SENT for sentence-final punctuation, and VV*/VH* for lexical verbs and HAVE. Tagging was done with this sheet. Because the NLTK training corpus uses the original PTB spelling (NNP, NNPS, PRP, PRP$, ., VB*), the Task 2 notebook maps each course tag one-to-one onto its PTB equivalent. The submitted file keeps both columns: tag_as_annotated and tag_ptb.

Conventions for cases specific to Sri Lankan news were applied consistently. Every part of a multi-word name is NP (Sri/NP Lanka/NP, Health/NP Promotion/NP Bureau/NP). Titles before a name are NP (Dr, Mr, MP). Rs. is tagged $, like US$ in the PTB, and % is NN. US meaning the United States is NP.

Tokens whose tag is fixed by the PTB rules (punctuation, brackets and quotes, the determiners the/a/an, the word to, plain numbers and the currency marker Rs.) were pre-filled by rule and checked

by the author. All other tokens, about 3,250 of the 4,237, were tagged by hand. Excel filters were used to tag repeated word forms together.

Quality control. Filter-based tagging gives every occurrence of a word form the same tag, which is wrong for context-dependent words (that as DT/IN/WDT, -ed forms as VVD/VVN, verbs after to). The tagged sheet was therefore checked in three review rounds. Rule-based checks (for example, a verb form directly after a modal or after to, an -ed form after was/has, that introducing a clause) and a comparison against how each word form is tagged in the PTB flagged suspicious tokens. The annotator re-examined each flagged token in context and corrected 304 tags (112 + 156 + 36). The final sheet passes the validation in the Task 2 notebook: every tag is legal, no token is untagged, and there are no punctuation or number slips.

The tagging was done by a single annotator, so no inter-annotator κ can be reported. Section 6.1 instead reports the agreement between the annotator and the PTB-trained model.

Example (sentence 50): ``/`` Some/DT media/NN institutions/NNS have/VHP misreported/VVN me/PP ,/, ''/'' he/PP said/VVD ./SENT

5. Task 3 – Evaluation Method

There are no gold-standard parse trees for the Daily Mirror sentences, so PARSEVAL cannot be computed on them directly. The lecturer confirmed that the aim is to test the sentences against the model and report the results obtained. Each sentence was tested at several levels (Table 5). At each level the question is how probable the sentence and the author's analysis are under the PTB model, and whether the model agrees with that analysis. All probabilities are normalised by sentence length, because raw log-probabilities fall with every extra word and would otherwise just rank sentences by length.

Level	Metric	What it shows
Tagging (HMM, Viterbi)	Accuracy; per-tag precision, recall, F1; Cohen's κ; confusion matrix	Agreement between the PTB-trained tagger and the human tags (the human tags are the reference)
Tag pattern (HMM transitions)	Tag-sequence perplexity = exp(−log $P(t_1 \ldots t_{n+1})/(n+1)$)	How typical the sentence's sequence of word categories is in WSJ English
Words + tags (HMM)	log P(w, t_human) per token; gap to the best (Viterbi) tagging	Likelihood of the sentence with the author's tags; a large gap means the model prefers another analysis
Words (HMM, forward algorithm)	Word perplexity = exp(−log $P(w_1 \ldots w_n)/n$); unknown-word rate	How surprising the vocabulary and style are compared with the WSJ
Structure (PCFG, CKY)	Full-parse coverage; number of fragments; log P(best parse) per token; PCFG perplexity	Whether the treebank grammar can build one tree over the sentence, and how probable that tree is
Structural robustness	Labelled-bracket F1 between the parse of the model's tags and the parse of the human tags	How much tagging disagreements change the syntactic analysis

Table 5. Metrics computed for every sentence.

5.1 Clustering rationale

To group the sentences by alignment, each sentence is described by four features: HMM log-probability per token (with the human tags), PCFG log-probability per token (including any fragment penalty), tagging accuracy, and unknown-word rate. The features are standardised to z-scores and

clustered with k-means (20 restarts). The number of clusters k is chosen by silhouette score over k = 2…5, and k = 3 (high / medium / low) is kept unless another k scores clearly higher (by more than 0.05). The number of PCFG fragments is reported for each sentence but not used as a clustering feature, because 148 of the 150 sentences receive a full parse, so it would only split off two outliers.

Clusters are named by their mean alignment score: the average of the z-scores, with unknown-word rate negated so that a higher score always means more PTB-like. The categories therefore have a value-based definition. A High-alignment sentence is one that is both probable under the model and agrees with it, whatever its topic. Section 6 describes each cluster by its average metrics and representative sentences, as the lecturer asked, rather than discussing every sentence.

6. Task 3 – Results

6.1 Tagging performance

Metric	Daily Mirror (150 sentences)	Held-out WSJ (Task 1)
Token accuracy	89.8%	94.7%
Accuracy on known words	91.7%	96.1%
Accuracy on unknown words	80.5%	82.8%
Unknown-word rate	16.9%	10.0%
Macro precision / recall / F1	0.746 / 0.752 / 0.744	–
Weighted F1	0.899	–
Cohen's κ (model vs human)	0.890	–
Sentences with no disagreement	12.0%	–

Table 6. Overall tagging performance of the PTB-trained HMM against the manual tags.

The model agrees with the manual tags on 89.8% of the 4,237 tokens (κ = 0.89, which is "almost perfect" agreement on the Landis–Koch scale). That is 4.9 points below the model's accuracy on held-out WSJ text. The drop is a measure of the domain shift from 1989 US financial news to 2026 Sri Lankan news. It comes mainly from known words used in unfamiliar ways (91.7% vs 96.1% on WSJ), not from unknown words. Although the unknown-word rate is 1.7 times higher than on WSJ, unknown-word accuracy (80.5%) is close to its WSJ value. Most unknown words are capitalised Sri Lankan names, which the shape signature UNK-CAP correctly sends to NNP. Only 12.0% of sentences have no disagreement at all, because in 28-token sentences a single disagreement is enough to break a sentence.

Tag	Precision	Recall	F1	Support
NN	0.824	0.853	0.838	578
IN	0.985	0.97	0.978	471
NNP	0.89	0.83	0.859	459
DT	0.994	0.967	0.98	362
NNS	0.945	0.939	0.942	347
JJ	0.683	0.82	0.745	255
,	1	1	1	254
VBD	0.85	0.667	0.747	153

Tag	Precision	Recall	F1	Support
.	1	1	1	152
CC	1	1	1	149
VBG	0.989	0.73	0.84	122
RB	0.938	0.876	0.906	121
VBN	0.705	0.835	0.764	103
TO	1	1	1	93
CD	0.989	1	0.995	91
VB	0.723	0.91	0.806	89
VBZ	0.809	0.974	0.884	78
PRP	0.986	1	0.993	71

Table 7. Per-tag precision, recall and F1 for the 18 most frequent tags (PTB spelling; human tags as reference).

Function-word and punctuation tags are almost perfect (DT, IN, CC, TO, CD, MD and punctuation all have F1 ≥ 0.97). Open-class and ambiguous tags are where the model and the annotator disagree: JJ (F1 0.75), VBD (0.75), VBN (0.76), VBP (0.69) and NN (0.84).

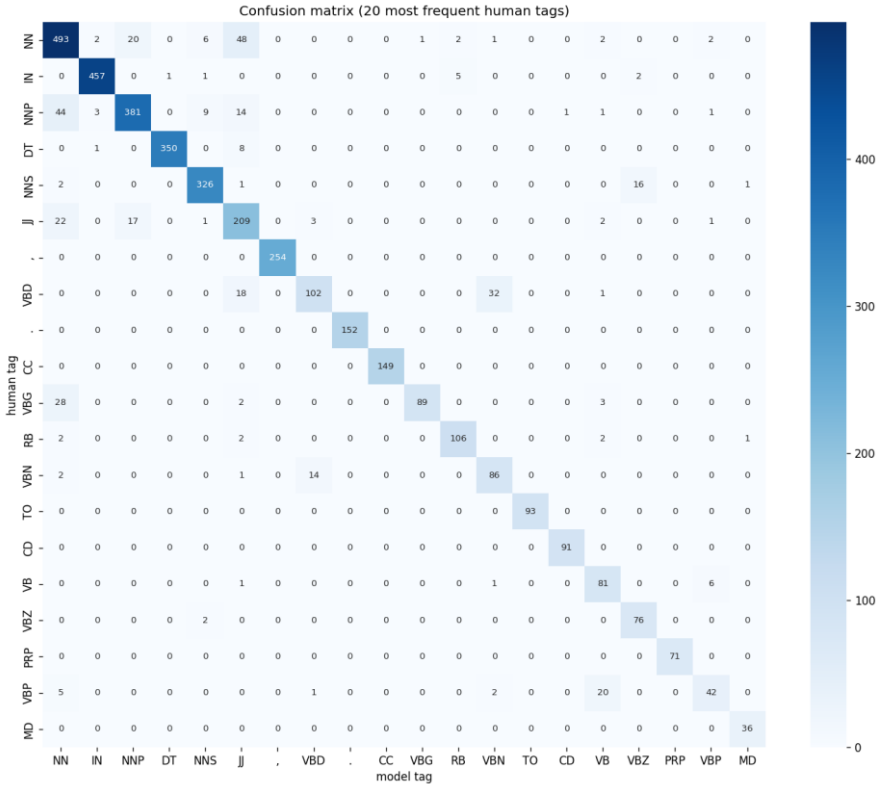


Figure 5. Confusion matrix for the 20 most frequent human tags (rows: human tag; columns: model tag).

Human	Model	Count	Example words
NN	JJ	48	preschool, illicit, simple, behaviour, white
NNP	NN	44	MW, Promotion, Campus, Consultant, Programme
VBD	VBN	32	conducted, seated, connected, aimed, involved
VBG	NN	28	learning, trafficking, understanding, teaching, training
JJ	NN	22	solar, official, crystal, consultant, sedentary
VBP	VB	20	learn, sleep, watch, receive, contribute

Human	Model	Count	Example words
NN	NNP	20	Sadewmi, Drug, Alongside, Iran, Energy
VBD	JJ	18	integrated, ground-mounted, extended, eye-related, smuggled
JJ	NNP	17	National, Northern, Medical, General, Western
NNS	VBZ	16	routes, girls, leaves, programmes, introduces

Table 8. The ten most frequent disagreements (PTB spelling).

The disagreements fall into three groups:

- **Genuinely ambiguous categories.** Noun vs adjective for words before a noun (preschool children, crystal methamphetamine, solar power). NNP vs NN for capitalised common nouns inside institution names (Health Promotion Bureau, Lyceum Campus, Programme). Past tense vs past participle for -ed forms (conducted, involved, seated). These are exactly the distinctions the PTB guidelines call difficult, and the annotator's choice is often defensible.
- **Model errors caused by the domain.** Local names that are ordinary words in the WSJ, and unit abbreviations such as MW (tagged NNP by the annotator, NN by the model). The bigram model also cannot see far enough to decide between, for example, VBP and VB after a coordinated subject.
- **Residual annotation slips.** Some disagreements, such as -ing nouns like trafficking and learning tagged as verbs (VBG), come from filter-based tagging of repeated word forms. These were largely reduced by the three review rounds but not eliminated. The model therefore also served as a useful second opinion on the manual annotation.

6.2 Likelihood and parsing performance

Measure	Mean	Median	Min	Max
Word perplexity (forward algorithm)	537	484	96	1847
Tag-sequence perplexity	11.5	10.8	6.2	27.6
PCFG log P(best parse) per token	-2.71	-2.69	-4.83	-1.91
Tagging gap per token (Viterbi − human)	0.51	0.43	0.00	1.91
Structure F1 (model-tag parse vs human-tag parse)	0.759	0.796	0.138	1.000

Table 9. Distribution of the likelihood and parsing measures over the 150 sentences.

The tag-level PCFG builds a single full tree for 98.7% of the sentences, as high as on held-out WSJ text (98.4%). At the level of word categories, Daily Mirror sentences are therefore well covered by WSJ phrase structure. Only two sentences needed a fragment parse: sentence 53 (4 fragments) and sentence 129 (8 fragments). Both are built around bracketed insertions and times ("the 2025 (2026) … (GCE O/L) Examination", "from 4.00 p.m. today (25) until 4.00 p.m. tomorrow (26)"), a pattern that is rare in the WSJ. The mean tag-sequence perplexity of 11.5 means that, on average, the next tag is about as uncertain as a choice among 11–12 equally likely tags. By contrast, word perplexity is high (median 484), reflecting the 12,000-word WSJ vocabulary rather than grammar. When the model's own tags are parsed instead of the human tags, the two trees share on average 76% of their labelled brackets. Tagging disagreements therefore do change the syntactic analysis, mostly in NP-internal structure and verb-phrase attachment.

6.3 Clusters of alignment

Silhouette scores are low and almost flat across k (k = 2: 0.245, k = 3: 0.269, k = 4: 0.27, k = 5: 0.262). The sentences therefore form a continuum of alignment rather than naturally separated groups, which is itself a finding. k = 3 was used, as it scores essentially as well as k = 4 and gives the most interpretable categories. The alignment score is not driven by sentence length (Pearson r = -0.064 between length and alignment score), which confirms that the per-token normalisation works.

Cluster	Sentences	Mean length	Tag accuracy	Unknown words	HMM log P / token	Median word perplexity	PCFG log P / token	Structure F1
High	38	25.4	93.2%	24.6%	-5.52	255	-2.40	0.85
Medium	69	29.7	93.1%	10.8%	-6.48	555	-2.84	0.80
Low	43	28.5	81.6%	18.5%	-7.18	620	-2.78	0.61

Table 10. Cluster profiles (cluster means; word perplexity is the cluster median).

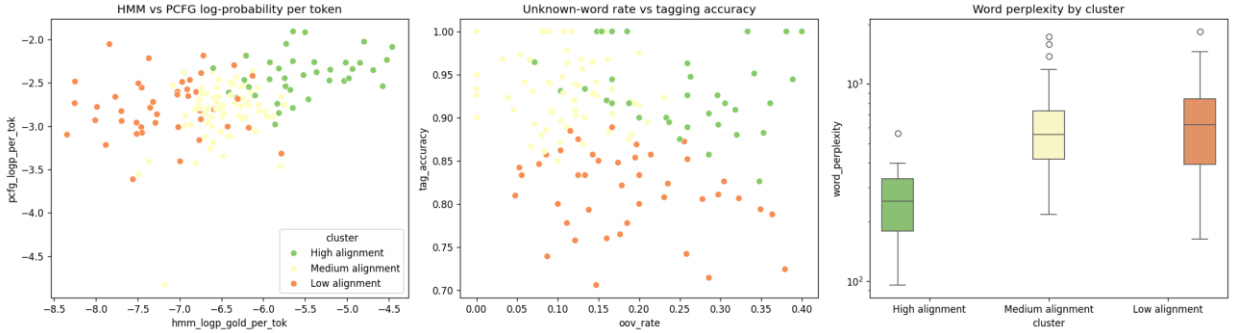


Figure 6. The three clusters in feature space: HMM vs PCFG log-probability, unknown-word rate vs tagging accuracy, and word perplexity by cluster.

High alignment. These 38 sentences have the highest tagging agreement (93.2%), the highest HMM and PCFG probabilities, and the lowest word perplexity. Surprisingly, they also have the highest unknown-word rate (24.6%). They are typically reported speech or name-heavy statements ("… he said", "In Galle, the warning covers the Neluwa, Nagoda, Niyagama and Thawalama Divisional Secretariat divisions"). Unknown names are handled well because runs of NNP tags are among the most probable tag bigrams in the WSJ (P(NNP | NNP) = 0.38), and "X said" is the WSJ's most typical sentence frame (34% of High-alignment sentences contain said, against 14% of Low-alignment ones). Unknown vocabulary on its own is therefore not what makes a sentence un-WSJ-like.

Medium alignment. The largest group (69 sentences) has equally high tagging agreement (93.1%) and few unknown words (10.8%), but lower probabilities. These are longer explanatory sentences with modal verbs, conditionals and embedded clauses ("For children between two and five years, screen time should not exceed 60 minutes a day…", "If the necessary conditions are met, the strait can be reopened…"). Their structure is grammatical and familiar, but less predictable token by token.

Low alignment. These 43 sentences combine the lowest tagging agreement (81.6%), the lowest HMM probability and the lowest structural agreement (F1 0.61). They are dominated by long coordinated noun-phrase lists and noun–noun compounds ("anatomy practical work, simple integrated clinical cases, clinical skills, research and critical appraisal…"), -ing nominalisations, and headline-like compression. In these constructions the NN/JJ/VBG choices are genuinely ambiguous.

Cluster	Type	Sent.	Align.	Tag acc.	Unknown	Sentence
High	typical	33	0.50	0.92	0.16	The government plans to increase solar power capacity connected to the national grid by a further 4,585.7 MW by 2029 , the Minister said .

Cluster	Type	Sent.	Align.	Tag acc.	Unknown	Sentence
High	typical	47	0.54	0.96	0.26	The verbal clash was witnessed between them when Mr Jayasekara said that there is a division within the government on the 22nd Amendment to the const…
High	typical	48	0.56	0.93	0.30	`` The government was hiding the fact. Mr Rathnayake revealed the truth , saying there is fractional division within the government , '' MP Jayasekar…
High	extreme	52	1.49	0.96	0.07	`` You said you did not work with divisions as a government , but there was a fractional division within the government , '' the MP said .
Medium	typical	7	0.16	0.89	0.11	For children between two and five years , screen time should not exceed 60 minutes a day and should preferably be limited to 30 to 45 minutes per day…
Medium	typical	101	0.16	0.93	0.11	`` If the necessary conditions are met , the strait can be reopened , a normal maritime passage restored within seven days , '' Araghchi told reporte…
Medium	typical	125	0.16	0.89	0.07	`` Will a deal be made with Iran that lets them rebuild and create a far greater country than it ever was before ? '' he said .
Medium	extreme	129	-1.08	1.00	0.08	The Level 3 (Red) warning is effective from 4.00 p.m. today (25) until 4.00 p.m. tomorrow (26) .
Low	typical	36	-0.61	0.81	0.28	The Sri Lanka Navy has seized illicit drugs estimated to be worth more than Rs. 55 billion through maritime and coastal anti-narcotics operations con…
Low	typical	80	-0.63	0.82	0.24	Through simple clinical scenarios , small-group discussions , models , prosected specimens , imaging , basic clinical observations and skills-lab act…
Low	typical	113	-0.66	0.86	0.14	But it fell apart a short time later when both sides resumed strikes .
Low	extreme	86	-1.69	0.74	0.26	Assessment includes anatomy practical work , simple integrated clinical cases , clinical skills , research and critical appraisal , Medical English ,…

Table 11. Representative sentences: the three closest to each cluster's median alignment score, plus the most extreme member of the cluster.

Alignment by source article. Grouping the scores by article (Table 12) shows that alignment follows register more than topic. The parliamentary exchange (reported speech) is the most PTB-like article. The NBRO landslide warning scores well despite having the highest unknown-word rate of the large articles. The sponsored pre-medical education feature is the least aligned large article: 19 of its 36 sentences fall in the Low cluster. It is written in a promotional, academic register with long lists of noun phrases that is far from newspaper financial prose. The republished BBC article, in international wire style, sits close to the average.

Article	Sent.	Tag acc.	Unknown	Mean align.	High / Med / Low
Screen use survey	24	91.3%	11.0%	0.18	2 / 16 / 6
Sathosa prices	4	89.7%	21.8%	0.10	2 / 1 / 1
Rooftop solar	6	87.7%	21.5%	-0.09	2 / 2 / 2
Navy drug routes	11	88.4%	21.5%	-0.13	3 / 4 / 4
Parliament exchange	7	93.8%	16.6%	0.68	6 / 1 / 0
O/L rescrutiny	3	82.4%	16.0%	-0.48	1 / 0 / 2
Pre-medical course (feature)	36	87.9%	14.1%	-0.26	1 / 16 / 19
Iran / Hormuz (BBC)	36	91.2%	14.0%	0.07	7 / 25 / 4
NBRO landslides	14	93.9%	26.5%	0.21	10 / 3 / 1
Gymnast Sadewmi	6	86.2%	30.2%	-0.21	4 / 0 / 2
Harry Potter play	3	80.1%	10.1%	-0.35	0 / 1 / 2

Table 12. Alignment by source article.

A low-alignment sentence in detail (sentence 86)

The least aligned sentence (alignment score -1.69, tag accuracy 74.2%, unknown words 25.8%) is shown word by word below. The disagreements cluster in the noun-phrase list, where several consecutive words could be nouns or adjectives and the bigram model has no way to see the phrase boundaries.

Word	Human	Model	Match
Assessment	JJ	NNP	✗
includes	VBZ	VBZ	✓
anatomy	NN	JJ	✗
practical	JJ	JJ	✓
work	NN	NN	✓
,	,	,	✓
simple	NN	JJ	✗
integrated	VBD	JJ	✗
clinical	JJ	JJ	✓
cases	NNS	NNS	✓
,	,	,	✓
clinical	JJ	JJ	✓
skills	NNS	NNS	✓
,	,	,	✓
research	NN	NN	✓
and	CC	CC	✓
critical	JJ	JJ	✓
appraisal	JJ	NN	✗
,	,	,	✓
Medical	JJ	NNP	✗
English	NNP	NNP	✓
,	,	,	✓
professionalism	NN	NN	✓
,	,	,	✓
teamwork	NN	NN	✓
and	CC	CC	✓
a	DT	DT	✓
final	JJ	JJ	✓
integrated	VBD	JJ	✗
assessment	JJ	NN	✗
.	.	.	✓

Table 13. Human vs model tags for the least aligned sentence (PTB spelling).

Best PCFG parse over the human tags: `(TOP (S (S (ADJP (JJ JJ) (S (VP (VBZ VBZ) (NP (NN NN) (JJ JJ) (NN NN)))))) (, ,) (NP (NP (NN NN)) (UCP (VP (VBD VBD) (NP (NP (JJ JJ) (NNS NNS)) (, ,) (NP (JJ JJ) (NNS NNS)))) (, ,) (UCP (NN NN) (CC CC) (NP (JJ JJ) (JJ JJ) (, ,) (NP (JJ JJ) (NNP NNP)))) (, ,) (NN NN) (, ,) (NN NN) (CC CC) (NP (DT DT) (JJ JJ)))) (VP (VBD VBD) (ADJP (JJ JJ))) (. .)))`

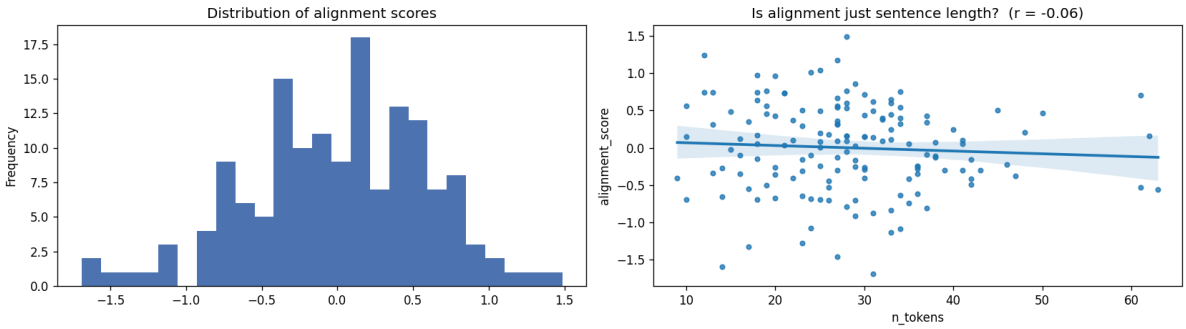


Figure 7. Distribution of alignment scores, and alignment against sentence length.

7. Discussion

7.1 Were the expected sources of misalignment confirmed?

- **Vocabulary: only partly confirmed.** The unknown-word rate is 16.9% (vs 10% on WSJ), but unknown words are tagged almost as accurately as on WSJ, and the highest-alignment cluster has the most unknown words. Sri Lankan names are unfamiliar words, but they occur in a familiar position: the NNP slot.
- **Proper-noun sequences: confirmed.** NNP → NN is the second most frequent disagreement: capitalised common nouns inside long institution names (Health Promotion Bureau, National Preschool Health Promotion Programme). This is a known grey area of the PTB guidelines.
- **Style and structure: confirmed for tagging, not for parsing.** VBD/VBN confusions in reduced relative clauses (a survey conducted across four provinces) and NN/JJ/VBG ambiguity in compounds and lists drive the Low cluster. However, the grammar still builds a complete tree for all but two sentences, so the treebank phrase structure itself transfers well. The problems are local category decisions, not missing constructions.
- **Annotation choices: confirmed.** The most frequent disagreement (NN vs JJ before a noun) is a genuine annotation ambiguity. In addition, comparing the manual tags with the model exposed several hundred annotation slips caused by filter-based tagging, which were corrected before the final evaluation. This shows why a model trained on a large, carefully annotated treebank is a useful quality check for new manual annotation.

7.2 Limitations

- The bigram HMM sees only one tag of context. A trigram model with interpolated smoothing, or a discriminative tagger, would typically gain 2–3 accuracy points and would resolve some VBD/VBN and NN/JJ cases.
- The NLTK sample is about 5% of the full WSJ treebank (3,914 vs about 49,000 sentences), so many rules and word–tag pairs are rare or unseen. A student raised this in the lecture, and the lecturer accepted it as a known limitation.
- The PCFG is unlexicalised and uses order-1 Markovisation with beam pruning. This trades accuracy for coverage and speed, and it cannot resolve attachment ambiguities that depend on the words.
- Without gold-standard trees for the Daily Mirror sentences, parsing performance is measured indirectly: by likelihood, coverage, fragmentation and agreement between the parses from human and model tags.
- The manual tags come from a single annotator. Despite three review rounds, some inconsistencies remain, so part of the measured disagreement is annotation noise rather than model error.

- The low, flat silhouette scores mean that the cluster boundaries are a convenient summary of a continuum, not sharp categories.

8. Conclusion

A bigram HMM POS tagger and a treebank-induced PCFG with a probabilistic CKY parser were built from the labelled Penn Treebank training data. On held-out WSJ text they reach 94.7% tagging accuracy and 75.2% labelled bracket F1. The full model statistics (tag-transition and emission probabilities and PCFG rule probabilities) are delivered as the PCFG parser data. The model was then used to test 150 manually tagged Daily Mirror sentences with tagging, likelihood and parsing metrics, and the sentences were grouped into clusters of alignment with the PTB.

On the Daily Mirror sentences the model agrees with the manual tags on 89.8% of tokens ($\kappa = 0.89$), and the PTB grammar builds a full parse for 98.7% of them. Sri Lankan news is therefore largely aligned with WSJ English at the level of phrase structure. It departs from the WSJ in local category decisions: noun–adjective modifiers, capitalised common nouns in names, and past-tense/participle forms. Register matters more than vocabulary: reported speech and name-heavy news align best, while promotional and academic prose with long noun-phrase lists aligns worst. A larger training treebank, a trigram or lexicalised model, and a second annotator would be the most effective improvements.

Appendix A – Submitted Files

File	Contents
Task1_POS_Tagger_and_PCFG_Model.ipynb	Task 1: training, model statistics, validation
model/	PCFG parser data: transition_matrix.csv, transition_counts.csv, emission_probs.csv, unknown_word_model.csv, tag_counts.csv, pcfg_grammar_lexicalised.txt, pcfg_rules.csv, pcfg_parser_grammar_cnf.txt, pos_pcfg_model.pkl, figures
Task2_Tagged_Sentences.ipynb	Task 2: tokenisation, tag validation, export
task2_150_tagged_sentences.xlsx / .csv / .txt	The 150 manually tagged sentences (course tags and PTB tags)
task2_tagging_sheet.xlsx	The tagging sheet (read by the Task 2 notebook)
task2_raw_sentences.txt, task2_sources.txt	Raw sentences and article sources
Task3_Test_Results.ipynb	Task 3: test results for every sentence, clustering
results/	Task3_test_results.xlsx, per_sentence_results.csv, overall and per-tag metrics, cluster profile and examples, figures